

UNITE-AUDIO: JOINT LEARNING OF CONTINUOUS TOKENIZATION AND LATENT FLOW MATCHING FOR TEXT-TO-AUDIO GENERATION

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ABSTRACT

Text-to-audio (TTA) generation aims to synthesize realistic audio that faithfully reflects natural-language descriptions. Most TTA systems adopt a two-stage latent paradigm: an audio tokenizer is optimized for reconstruction and then frozen, after which a generative model is trained in the resulting latent space. However, reconstruction-oriented representations may be suboptimal for generation, motivating joint representation and generative learning. To this end, we introduce **Unite-Audio**, to our knowledge, is the **first** to jointly learn continuous audio representations and latent flow matching for TTA. By coupling reconstruction with self-supervised generative prediction, Unite-Audio allows the generative objective to directly shape the latent space rather than treating it as a fixed intermediate representation. We further employ Flow-GRPO post-training to improve text-conditioned generation. Experiments show competitive TTA performance with a compact latent flow model, while ablation studies confirm the benefit of jointly learning the audio representation and generative model. Audio samples are available at <https://runwushi.github.io/Unite-Audio/>.

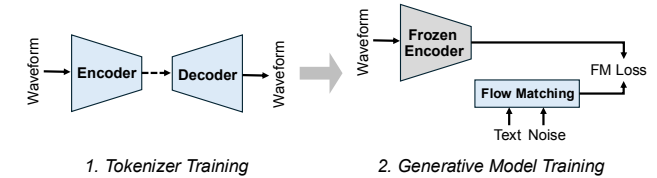
Index Terms— Text-to-audio generation, Flow matching, Audio representation learning, Latent generative models

1. INTRODUCTION

Recent text-to-audio (TTA) systems have made substantial progress in generating diverse acoustic scenes and sound events from open-ended textual descriptions. Modern TTA systems predominantly adopt latent generative modeling, where high-dimensional audio signals are compressed into compact representations and a conditional diffusion or flow model is trained to generate within this latent space [1, 2, 3].

However, in this two-stage paradigm, the audio tokenizer is typically optimized for reconstruction independently of the downstream generative objective. As a result, the learned latent space, while effective for faithful decoding, may not necessarily be the most suitable representation for generative modeling. To improve the generative properties of reconstruction-oriented latents, recent speech generation systems have incorporated semantic supervision from pretrained language or automatic speech recognition (ASR) models into

Separate Training (Two Stages Pretrain)



Joint Training (Ours, Single Stage Pretrain)

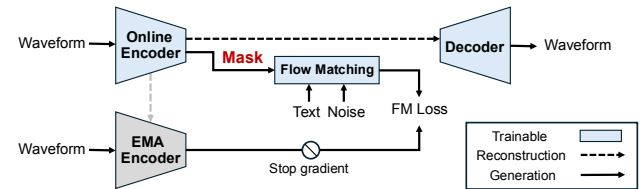


Fig. 1. Comparison between conventional separate tokenizer-generator training and our joint single-stage training.

tokenizer training. For example, Qwen-Audio-3.0-Gen leverages a pretrained language model [4], while Ming-UniAudio introduces semantic knowledge from a Whisper encoder to shape their continuous audio representations [5].

Beyond aligning tokenizers with external semantic models, recent studies in visual generation have explored shaping the latent space directly through self-supervised and generative objectives. MAETok introduces masked modeling into tokenizer training, showing that more structured latent representations can facilitate downstream generative modeling while maintaining reconstruction fidelity [6]. More recent approaches further couple representation learning with latent generative modeling. Unified Latents regularizes learned representations with a diffusion prior [7], while UNITE jointly optimizes tokenization and latent denoising in a single-stage framework, allowing the generative objective to directly participate in shaping the latent space [8]. Meanwhile, recent advances in visual generation have shown that self-supervised representations can provide strong priors for generative modeling. Methods such as REPA [9], RAE [10], and ReDi [11] exploit pretrained representations or jointly model semantic features to improve generative performance. These studies suggest that generation can benefit not only from better optimization of the generative model, but also from a latent space

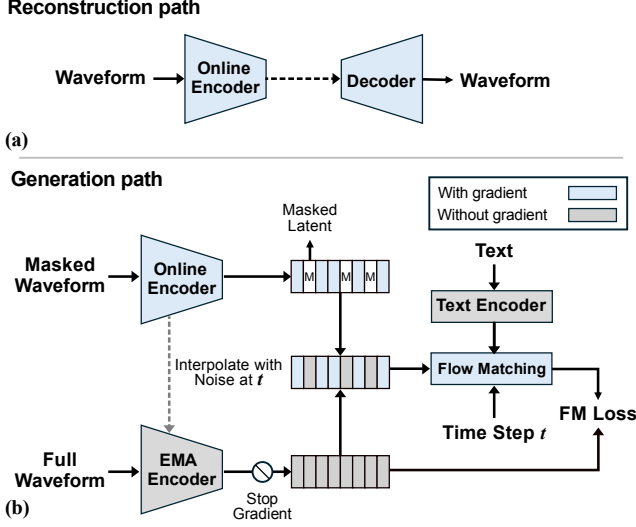


Fig. 2. Overview of the proposed framework: (a) the reconstruction path and (b) the generation path. Blue denotes regions with gradient propagation, while gray denotes regions without gradient propagation.

endowed with stronger representation structure.

Building on these two complementary directions, we introduce **Unite-Audio**, which, to the best of our knowledge, is the first TTA framework to jointly optimize continuous audio tokenization and latent generative modeling. Unite-Audio further incorporates self-supervised prediction into latent representation learning and learns its latent space entirely from scratch, without relying on external pretrained models for representation alignment. Specifically, the online encoder observes a partial waveform and produces latent representations that are noised and fed to the latent flow model, while an exponential-moving-average (EMA) encoder observes the full waveform and provides the full latent target. By predicting the full EMA representation, the flow-matching objective can propagate through the online representations and directly shape the encoder, while reconstruction preserves the information required for faithful decoding. We additionally interleave full-generation training from Gaussian noise to match the inference setting. An overview of the proposed framework is illustrated in Figure 1. Experimental results and ablation studies demonstrate the effectiveness of these designs for learning a generation-friendly latent space, with Unite-Audio achieving competitive TTA performance despite its compact model size.

2. UNITE-AUDIO FRAMEWORK

2.1. Overview

An overview of the training framework of Unite-Audio is illustrated in Figure 2. Let $x \in \mathbb{R}^T$ denote an audio waveform and c its caption. Unite-Audio consists of an online encoder E_θ , a decoder D_ϕ , an EMA encoder $\bar{E}_\theta$, and a text-

conditioned latent flow model F_ψ . During training, a fraction p_{rec} of each batch follows the reconstruction path, while the remaining $1 - p_{\text{rec}}$ follows the generation path; both paths propagate gradients to E_θ and jointly shape the latent space.

For reconstruction, the full waveform is encoded as $z_{\text{rec}} = E_\theta(x)$ and decoded by D_ϕ . For generation, the online encoder observes only partial waveform context, while the EMA encoder observes the full waveform and provides the stop-gradient target $z^* = \text{sg}(\bar{E}_\theta(x))$. The partial online representations are incorporated into the flow-model input while retaining their gradient connection to E_θ , allowing the generative objective to directly participate in representation learning.

2.2. Partial-Context Flow Matching

For the generation path, we divide each waveform into non-overlapping 2-second temporal buckets and independently construct partial context within each bucket. Specifically, the context ratio r_{ctx} of each bucket is independently sampled between 0 and 30%, where the context refers to the retained (unmasked) waveform regions. Given the resulting contextual waveform x_{ctx} , the online encoder produces $z^{\text{ctx}} = E_\theta(x_{\text{ctx}})$, while the EMA encoder observes the full waveform and provides the full-view target z^* .

To reduce the discrepancy between partial-context training and generation from noise, we noise the contextual representation at the same flow time t as the EMA trajectory. Given $\epsilon \sim \mathcal{N}(0, I)$, we define $z_t^* = (1 - t)\epsilon + tz^*$ and $z_t^{\text{ctx}} = (1 - t)\epsilon + tz^{\text{ctx}}$. Let $\mathcal{C}$ denote the contextual positions. The complete flow input is constructed as

$$\tilde{z}_{t,i} = \begin{cases} z_{t,i}^{\text{ctx}}, & i \in \mathcal{C}, \\ z_{t,i}^*, & i \notin \mathcal{C}. \end{cases} \quad (1)$$

Thus, F_ψ always receives a complete noisy latent sequence, while only the contextual representations retain a gradient path to E_θ . Unlike conventional masked self-supervised learning, using clean online context together with noisy flow states leads to a mismatch and degrades generation performance. We therefore noise the online context to the same flow time t .

We adopt an x -prediction flow-matching objective, using the clean full-view EMA representation as the target:

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{t,\epsilon} \left[\frac{1}{|\mathcal{C}|} \sum_{i \notin \mathcal{C}} \|F_\psi(\tilde{z}_t, c, t)_i - z_i^*\|_2^2 \right], \quad (2)$$

where the loss is evaluated over the target regions complementary to the online context. Following UNITE [8], we use a shifted flow-time schedule that emphasizes high-noise states. Through the contextual positions in z_t^{ctx} , $\mathcal{L}_{\text{FM}}$ can propagate to the online encoder.

2.3. Training Objectives and Schedule

The reconstruction objective follows recent audio autoencoder training recipes [12, 13]. We combine global spectral

reconstruction, local multi-resolution spectral reconstruction, and adversarial learning. The global STFT loss $\mathcal{L}_{\text{GSTFT}}$ is computed over the full valid waveform to preserve global spectral structure. The phase-aware MRSTFT loss $\mathcal{L}_{\text{MRSTFT}}$ is computed on random 2-second segments using seven dyadic FFT sizes from 32 to 2048. We further use a GAN loss $\mathcal{L}_{\text{GAN}}$ with multi-view STFT/PQMF discriminators and a feature-matching loss $\mathcal{L}_{\text{feat}}$ to improve perceptual realism and stabilize adversarial training [13]. The reconstruction objective is defined as

$$\mathcal{L}_{\text{rec}} = \lambda_{\text{G}}\mathcal{L}_{\text{GSTFT}} + \lambda_{\text{M}}\mathcal{L}_{\text{MRSTFT}} + \lambda_{\text{GAN}}\mathcal{L}_{\text{GAN}} + \lambda_{\text{F}}\mathcal{L}_{\text{feat}}. \quad (3)$$

Here, λ_{G} , λ_{M} , λ_{GAN} , and λ_{F} control the corresponding loss contributions. The discriminator is optimized separately using $\mathcal{L}_{\text{D}}$.

For latent generation, we use the x -prediction flow-matching loss $\mathcal{L}_{\text{FM}}$ defined in the previous section. With p_{rec} denoting the probability of reconstruction routing, the joint objective is

$$\mathcal{L}_{\text{total}} = p_{\text{rec}}\mathcal{L}_{\text{rec}} + (1 - p_{\text{rec}})\mathcal{L}_{\text{FM}}. \quad (4)$$

The reconstruction objective maintains acoustic fidelity and decodability, while flow matching provides predictive and generative supervision to the latent representation.

Stage I: Joint latent training. We jointly optimize E_{θ} , D_{ϕ} , and F_{ψ} using $\mathcal{L}_{\text{total}}$ from scratch. Partial-context flow matching allows $\mathcal{L}_{\text{FM}}$ to propagate to the online encoder, while full-generation examples expose F_{ψ} to generation directly from noise. The EMA encoder $\bar{E}_{\theta}$ receives no gradients and is updated only through exponential moving averaging.

Stage II: Full-generation training. We freeze E_{θ} , D_{ϕ} , and $\bar{E}_{\theta}$ and optimize only F_{ψ} using $\mathcal{L}_{\text{FM}}$. Reconstruction and partial-context training are removed, specializing the flow model to text-conditioned generation from Gaussian noise while keeping the latent representation fixed.

Stage III: Flow-GRPO post-training. Finally, we further optimize F_{ψ} using Flow-GRPO [14], while keeping the encoder and decoder fixed. For each caption, multiple generations are sampled to form a group. The composite reward is defined as $R_i = \lambda_{\text{CLAP}}z_g(r_i^{\text{CLAP}}) + \lambda_{\text{PaSST}}z_g(-r_i^{\text{PaSST}})$, and the group-relative advantage is $A_i = z_g(R_i)$, where $z_g(\cdot)$ denotes within-group normalization. Here, r_i^{CLAP} is the CLAP similarity, while r_i^{PaSST} is the PaSST KL divergence between the generated audio and its paired reference audio from the training set. The policy is optimized using the clipped Flow-GRPO objective:

$$\begin{aligned} \mathcal{L}_{\text{GRPO}} = & \mathbb{E}_{i,t}[-\min(\rho_{i,t}A_i, \text{clip}(\rho_{i,t}, 1 - \epsilon, 1 + \epsilon)A_i)] \\ & + \beta D_{\text{KL}}(\pi_{\psi} \parallel \pi_{\text{ref}}), \end{aligned} \quad (5)$$

where $\rho_{i,t}$ is the transition likelihood ratio between the current and rollout policies, and π_{ref} is the frozen Stage-II flow model.

3. EXPERIMENTS

3.1. Dataset

Unite-Audio uses three audio-caption datasets across the training stages. Stage I uses 1,729,107 AudioSet examples [15] and 246,324 WavCaps examples [16], with all AudioCaps test-set entries explicitly removed from the AudioSet-derived data to prevent evaluation leakage. Stage II and Stage III use the AudioCaps training set [17]. We retain WavCaps samples shorter than 20 seconds and resample all audio to 16 kHz with a maximum duration of 20 seconds. Before encoding, waveforms are scaled by a fixed factor of 8, and the resulting latent representations are RMS-normalized to a fixed scale of 1.0, such that the flow model operates on a unit-RMS latent space.

3.2. Model Architecture

Unite-Audio consists of a continuous audio tokenizer and a text-conditioned latent flow model. The tokenizer operates on 160-sample waveform patches and produces 64-dimensional latent representations at 25 Hz. The encoder and decoder each contain 12 same-resolution Transformer blocks, followed by two downsampling and upsampling blocks, respectively. The latent flow model follows the multimodal diffusion transformer (MMDiT) architecture [18], where audio and text representations interact through joint attention blocks. The flow model uses a hidden dimension of 512 and 18 Transformer layers, including 11 fused audio-text MMDiT blocks. Text conditioning is provided by a frozen FLAN-T5-Large encoder [19]. The trainable latent flow model contains 117.2M parameters, excluding the tokenizer and frozen text encoder.

3.3. Implementation Details

For Stage I, we train for 240k iterations with a batch size of 24 using AdamW with a learning rate of 1×10^{-4} . The EMA decay is set to 0.999. The context ratio r_{ctx} is sampled from 0 to 30%, and the reconstruction sampling probability p_{rec} is set to 0.5. The reconstruction loss weights are $\lambda_{\text{G}} = 1.0$, $\lambda_{\text{M}} = 1.0$, $\lambda_{\text{GAN}} = 0.1$, and $\lambda_{\text{F}} = 5.0$. Adversarial training starts after 10k steps. Text conditioning is dropped with a probability of 10% for classifier-free guidance (CFG).

For Stage II, we use AdamW with a learning rate of 5×10^{-5} , weight decay of 0.01, and batch size 64. For Stage III, we use a learning rate of 1×10^{-6} and group size 4. Reward weights are $\lambda_{\text{CLAP}} = 2/3$ and $\lambda_{\text{PaSST}} = 1/3$, with $\epsilon = 10^{-4}$ and $\beta = 0.01$. For Unite-Audio, the CFG scale is 5.0 for 32-step sampling and 2.0 for lower-NFE settings. All experiments use two NVIDIA H800 GPUs.

3.4. Evaluation

We follow the TangoFlux evaluation protocol [3] on the 886-sample AudioCaps test split. We report OpenL3-based Fréchet Distance (FD), PaSST-based KL divergence (KL), CLAP similarity, and Inception Score (IS). FD measures the distributional similarity between generated and reference

Table 1. Comparison under the TangoFlux AudioCaps-886 protocol. Params. counts only the parameters of the latent generative model, excluding the audio encoder–decoder and text encoder. [‡] Results are directly taken from the TangoFlux benchmark [3]; [†] denotes our evaluation of official checkpoints; * denotes FD_{16k} results computed by us. S1, S2, and S3 denote the three training stages of Unite-Audio. Best results are bold and underlined, and second-best results are bold.

Model	Params.	Text Cond.	Vocoder	NFE	FD _{OpenL3} ↓	FD _{16k} * ↓	KL _{PaSST} ↓	CLAP ↑	IS ↑	RTF ↓
AudioLDM2-Large [‡] [2]	712M	T5 + CLAP	✓	200	108.3	40.03	1.81	0.419	7.90	2.589
Stable Audio Open [‡] [20]	1056M	T5	×	100	89.2	74.01	2.58	0.291	9.90	0.413
Tango2 [‡] [21]	866M	T5	✓	200	108.4	45.45	1.11	0.447	9.00	0.781
TangoFlux-Base [‡] [3]	515M	T5	×	50	80.2	49.76	1.22	0.431	11.70	0.195
TangoFlux-RL [‡] [3]	515M	T5	×	50	75.1	52.28	1.15	0.480	12.20	0.195
MeanAudio [†] [22]	120M	T5 + CLAP	✓	25	121.79	88.11	1.214	0.455	10.58	0.011
EzAudio [†] [23]	875M	T5 + CLAP	×	50	38.26	41.89	1.179	0.496	9.77	0.305
GenAU [†] [24]	1250M	T5 + CLAP	✓	200	68.92	31.94	1.379	0.473	10.38	0.883
Unite-Audio S1 (CFG=5.0)	117M	T5	×	32	76.04	38.61	1.598	0.457	10.24	0.062
Unite-Audio S2 (CFG=5.0)	117M	T5	×	32	73.46	33.41	1.205	0.502	10.41	0.062
Unite-Audio S3 (CFG=5.0)	117M	T5	×	32	84.76	43.25	1.107	0.534	11.34	0.062
Unite-Audio S3 (CFG=2.0)	117M	T5	×	16	83.22	39.38	1.057	0.532	11.04	0.031
Unite-Audio S3 (CFG=2.0)	117M	T5	×	8	84.88	42.41	1.079	0.530	10.85	0.016
Unite-Audio S3 (CFG=2.0)	117M	T5	×	4	88.91	48.86	1.175	0.512	9.73	0.009

audio, KL compares their PaSST-predicted event distributions, CLAP measures text–audio semantic alignment, and IS reflects sample diversity and specificity [25, 26, 27]. We additionally report FD_{16k}, where both generated and reference audio are resampled to 16 kHz before evaluation. Other metrics follow the same evaluation models and settings as TangoFlux. RTF is averaged over ten runs of 10-second generation on a single NVIDIA H800 GPU.

4. RESULTS AND ABLATION STUDY

4.1. Performance

Table 1 compares Unite-Audio with existing TTA systems on the AudioCaps-886 benchmark. With a 117M-parameter latent flow model, Unite-Audio achieves competitive overall performance. Stage II already shows strong generation before reward-based post-training, while Flow-GRPO further improves semantic alignment and event consistency. The final model achieves the highest CLAP score of 0.534 and the lowest KL_{PaSST} of 1.057 among the compared systems, together with a competitive IS. Although FD degrades after Flow-GRPO, Stage II provides a better trade-off in distributional fidelity. Unite-Audio also remains robust under fewer sampling steps, with only modest degradation at 8 NFEs.

4.2. Necessity of Noisy Partial-Context Flow Matching

We compare three Stage-I checkpoints at 200k iterations, evaluated using EMA weights on the same 886-sample AudioCaps test set, to examine how different context-learning strategies affect generation and the learned encoder representations. *No context* removes the online representation from flow matching, such that $\mathcal{L}_{\text{FM}}$ has no direct gradient path to the encoder. *Clean context* restores this gradient path but introduces clean latent information into otherwise noisy flow states, creating a mismatch between the context and the current generative state. Our *noisy context* instead places the

Table 2. Evaluation of latent representations learned with different context-learning strategies at 200k iterations.

Method	FD↓	KL↓	IS↑	CLAP↑	Rank↑	Acc.↑	Mel↓	SI-SDR↑
No context	79.87	1.811	9.68	0.444	8.44	0.36	0.96	4.66
Clean context	82.38	1.871	8.60	0.434	8.06	0.37	0.95	4.85
Noisy (ours)	80.19	1.669	10.37	0.454	8.90	0.41	0.94	4.72

online representation at the same flow time as the generative state.

Beyond generation metrics, we report effective rank to quantify latent dimensional utilization, defined as $r_{\text{eff}} = \exp(-\sum_i p_i \log p_i)$, where $p_i = \sigma_i^2 / \sum_j \sigma_j^2$ denotes the normalized singular-value energy. Semantic separability is further evaluated on ESC-50 [28] using a linear probe, with classification accuracy reported as Acc. Following Liu et al. [29], encoder-decoder reconstruction quality is evaluated using multi-scale Mel distance and scale-invariant signal-to-distortion ratio (SI-SDR). As shown in Table 2, noisy context performs best on most generation and representation metrics, achieves the lowest Mel distance, and maintains comparable SI-SDR, indicating improved representation learning without compromising reconstruction quality.

5. CONCLUSION

We introduced **Unite-Audio**, a framework that jointly learns continuous audio tokenization and latent flow matching for TTA generation. Noise-matched partial-context flow matching allows the generative objective to directly shape the encoder while avoiding clean-context information unavailable at inference, and full-generation training further aligns learning with generation from noise. Flow-GRPO post-training further improves text-conditioned generation. Experiments show that Unite-Audio achieves strong semantic fidelity with a compact 117M-parameter latent flow model, while retaining competitive performance under few-step sampling.

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